

Video Sports Analytics

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Abstract: Video sport analytics is a rapidly evolving field, offering organizations and players advanced technologies to enhance understanding and performance, and to deepen our insights into the athletic body. In fast-paced sports such as football and basketball, capturing finer details during live matches is challenging. Human observers struggle to simultaneously analyze player performances, opponent tactics, and other critical aspects of the game. Post-game footage review is crucial for identifying strengths, weaknesses, and areas for improvement, but traditional video analysis methods are plagued by significant time and cost constraints.

These challenges can be significantly addressed using computer vision. Existing methods can accurately recognize player actions and track them, allowing for deeper analysis of individual performances. Automating highlight reels saves time and ensures that the most important events are captured. With enough data, temporal analysis—studying events over time—can reveal important patterns. Collectively, these advancements enhance our understanding of athletes and teams while saving both time and money.

As we will see in detail below, analyzing videos in sports has provided a wide range of applications, such as recognizing player positions, content extraction, summarization, high-level analysis of players and actions, and other fundamental elements required for sport analysis.

1.Introduction

Nowadays, it is hard to imagine modern sports and exercise science in general without mentioning the contribution of technology and data gathering. However until the 1990s, video analysis in sports was not that extensive. First in the US college and professional teams were the early adopters of the technology but the landscape was very different from what it is today. Coaches and players would meticulously go over replays and full game footage, trying to get the edge over their opponents making the process incredibly time-consuming and labor-intensive, only reserved to teams with enough resources to hire experts on the field.

By the early 2000s, advancements in digital technology such as higher quality video capturing equipment, sophisticated software and increased computational power began

to transform video analysis. The integration of computer vision and machine learning marked a significant leap forward. Now, it is also being permanently added as a feature on match days to aid referees in major sports like the NFL and football. However, these advancements have not come without criticism from both fans and players. In recent years, there have been extensive discussions about the negative aspects of video analysis. The most notable example is the Video Assistant Referee (VAR), which was implemented in major football tournaments starting from the 2018 World Cup. Whether the criticism was fair or not, it highlights that people often react strongly to automation in sports. Therefore, major breakthroughs in these technologies might benefit from a gradual, carefully managed introduction before reaching the end consumer.

These discussions should happen, but at the same time, they should not hinder the development of the field, as the advantages far outweigh the disadvantages. Some of the most important benefits of automation in video sport analysis are:

- **Enhanced Accuracy:** Automation will reduce human error, ensuring more precise analysis of game footage.
- **Time Efficiency:** The analysis is sped up and as a result organizations and athletes will have data available in less time, which enables quicker decision making and insight generation.
- **Real-Time Analysis:** Real-Time processing is crucial for coaches to analyze match content live and make immediate strategic adjustments.
- **Data Collection:** We can gather extensive and detailed information on player actions, game events etc. while generating more accurate and insightful results.
- **Reduced cost:** One of the main objects for many of these technologies is to democratize them and let development and low budget teams have access to the same level of analysis world class teams have.
- **Scalability:** Accurate models developed for one sport can be extended to others.
- **Objective Decision-Making:** Technologies like VAR allow referees to make evidence-based decisions, while fans benefit from visual aids and interactive features.

Those are just some of the most important advantages. This literature review aims to explain some of the technologies that already exist, their applications and challenges. To succeed in this, I will mention two important papers and their impact on the field.

2.PassNet: Automated Sports Analysis Using Computer Vision

Nowadays sport analytics are developing rapidly, thanks to the availability of data that describe spatio-temporal events during matches we can have more in depth analysis. PassNet offers an automatic annotation system for passes, which are the most frequent event in a football match (~50%). The authors mention the importance of this approach, as no other automatic pass detection models exist. This system detects passes combining three models:

► Feature extraction

A crucial step in the process, where relevant characteristics or patterns are identified and extracted from raw data. ResNet18, an architecture within the family of Residual Networks is used due to its ability to handle complex visual data efficiently.

► Sequence Classification

They employed Bidirectional-LSTM for classification of temporal sequences (e.g. passes)

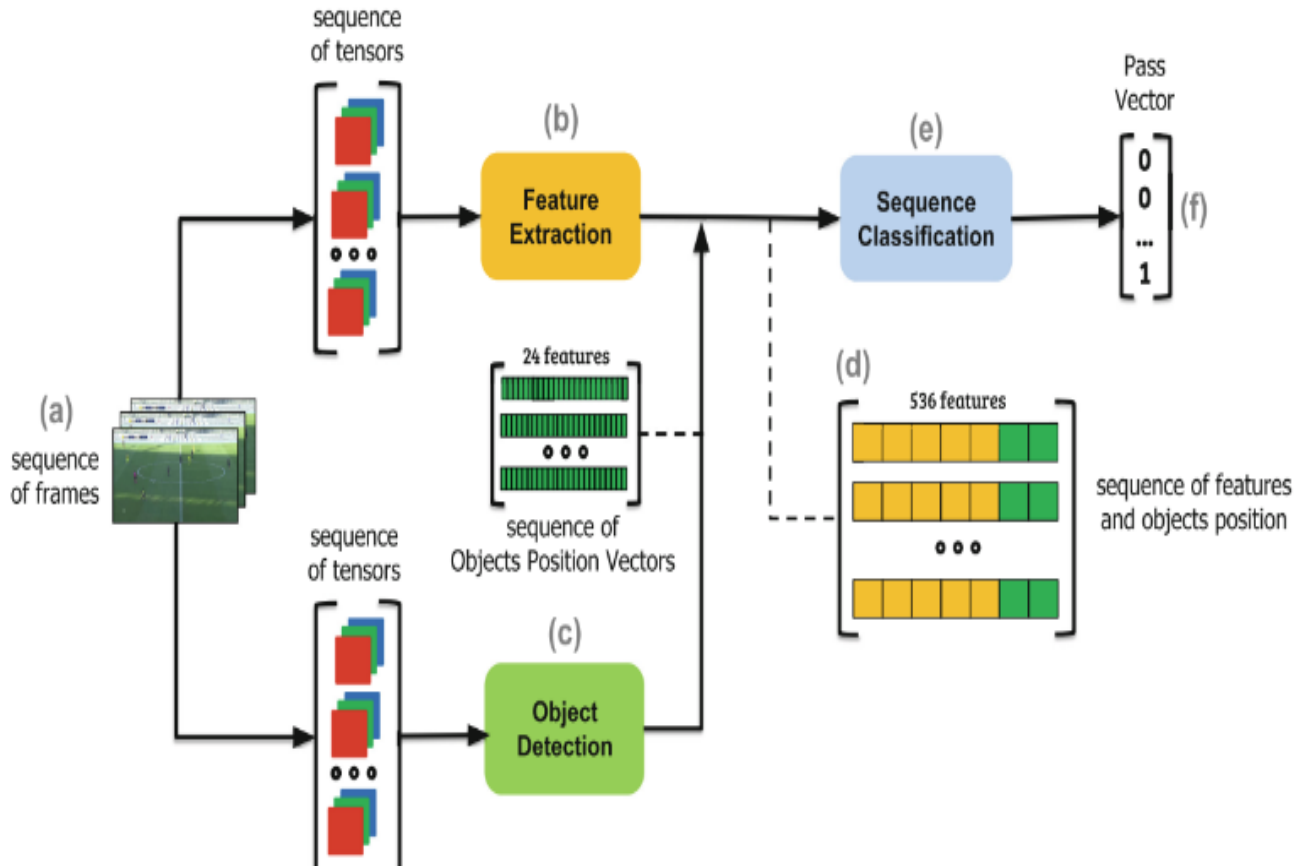
► Object Detection

Uses YOLOv3, a state-of-the-art object detection algorithm which was pretrained on the COCO dataset. The COCO dataset is a large-scale object detection dataset containing over 80 object categories, including ball, players, and other sports-related objects.

2.1 The architecture of PassNet

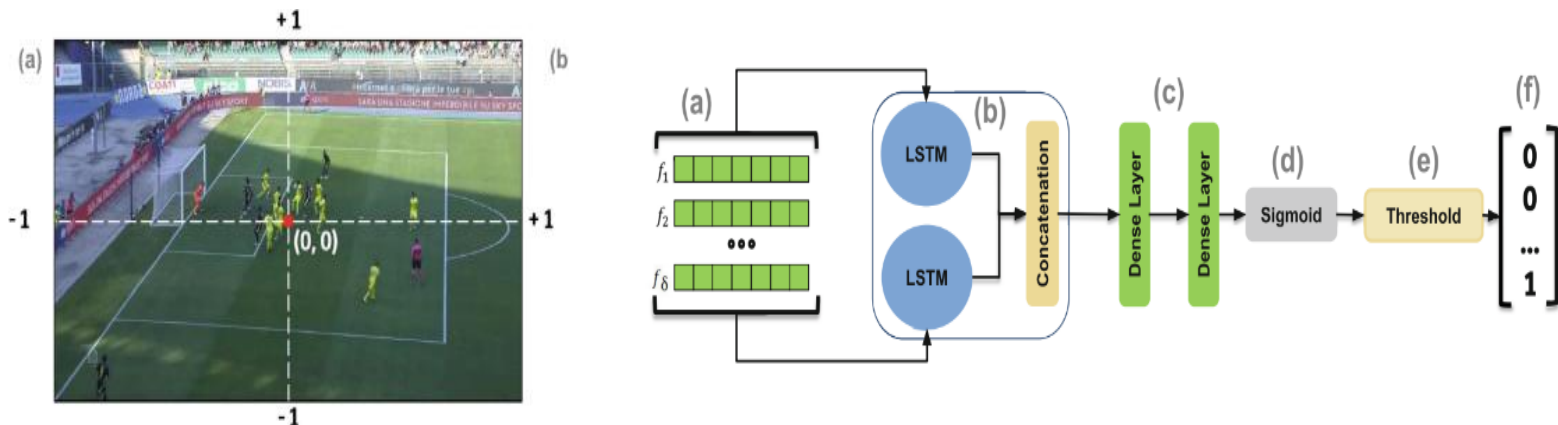
A sequence of frames is provided in input with dimensions 3 x 352 x 240. This denotes a RGB image (3 channels) with a resolution of 352 x 240, and each one is passed through a feature extraction module which uses ResNet18, resulting in a sequence of 512 features per frame. These features contain high-level information, such as textures, signs, and patterns. An object detection module using YOLOv3 produces bounding boxes of the objects found, creating a sequence of 24 features that describe the position of the ball and the closest players.

The 512 features from the feature extraction module are combined with the 24 features from the object detection module. The result is a combined feature vector of 536 features per frame that indicates if a frame is part of a pass sequence or not.



2.2 The sequence classification

The combination of the features that resulted in the sequence of vectors with 536 features, is provided as input to the sequence classification module. Using a sliding window δ the sequence is split into sub-sequences of length δ . Each sub-sequence is fed into the Bidirectional-LSTM that processes them in both forward and backward directions and the output is a vector of δ values, where each value is either 1 (indicating the frame is part of a pass sequence) or 0 (indicating the frame is not part of a pass sequence).

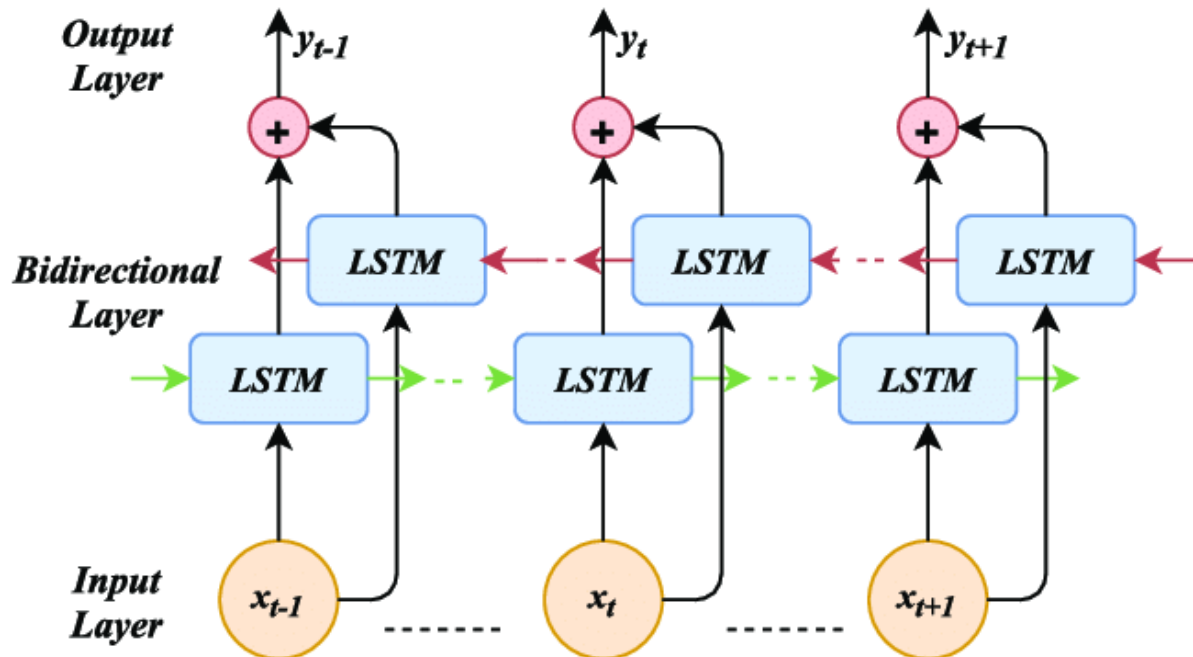


a)The coordinates were normalized in the $[-1,1]$ range

b)The bidirectional-LSTM process the sub-sequence from f_1 to f_δ and from f_δ to f_1 (first->last, last->first). The output is provided to two dense layers with ReLu activation functions(dropout=0.5). A sigmoid activation function combined with an activation threshold, will give us the output of δ values(0->no pass, 1->pass)

2.3 The Bidirectional-LSTM

Bidirectional Long Short-Term Memory (LSTM) is a critical component in sequence classification. It is a special kind of Recurrent Neural Network (RNN) architecture which is designed to model sequences and their long-range dependencies more precisely than RNNs. LSTMs also address issues of vanishing gradients found in traditional RNNs. In the context of this paper, by analyzing the sequence bidirectionally, the network can understand the influence of both preceding and succeeding frames on the current frame, improving the accuracy of pass sequence detection. Bi-LSTMs are also flexible to handle different length-sequences efficiently while their versatility and transfer learning capabilities makes them an excellent candidate for achieving the goals of this paper, which we will discuss further in section [2.5].



The architecture of a Bi-LSTM: We provide the input sequence of frames(input layer), the forward and backward LSTM processes happens simultaneously, capturing dependencies from both past and future contexts. As the LSTM processes each frame, it generates

hidden states at each time step. These hidden states encapsulate the learned information up to that point in the sequence. At each time step, the outputs from the forward and backward LSTMs are concatenated and then further processed (as mentioned in 2.2). The processed outputs are then used to make predictions.

2.4 The dataset-application

Video streams from Serie A were processed, with the resolution reduced to 352x240 and the frame rate to 5 fps. The authors also implemented a web application with a user interface where they can manually annotate the starting and ending times of a pass.

Passes were manually annotated using the UI of the web application, resulting in a binary vector (e.g. [0011111000]).

The screenshot shows a web application titled "Events Tagging Dashboard". At the top, there is a navigation bar with a "Matches" dropdown menu labeled (a). Below this, the main content area is divided into two sections. On the left, labeled (c), is a video player titled "Sassuolo vs. Internazionale 1° Tempo (Pass)". The video shows a football match in progress. Below the video player are buttons for "Pass Start" and "Pass End". On the right, labeled (b), is a table titled "Last Update: 21/10/2019 - 17:10:50". The table lists pass events with columns for Id, Player, Team, Timestamp, Pass Start, and Pass End. Below the table is a section for "Events Type" with radio buttons for "Pass" and "Shot".

Id	Player	Team	Timestamp	Pass Start	Pass End
217028173	F. Magnanelli	Sassuolo	0.77	0.12	1.63
217028175	F. Acerbi	Sassuolo	3.07	2.62	5.6
217028162	R. Gagliardini	Internazionale	5.7	5.7	6.83
217028184	F. Magnanelli	Sassuolo	7.31	7.01	7.78
217028186	D. Berardi	Sassuolo	8.11	7.86	8.94
217028198	F. Magnanelli	Sassuolo	15.75	15.65	16.54
217028200	Poi Lirola	Sassuolo	18.39	17.99	20.39
217028167	M.	Internazionale	16.03	15.48	17.54

The user can load a match using the appropriate dropdown (a). On the right side (b), a table shows all the pass events of the match and related information. On the left side (c), the interface shows the video and buttons to start and pause it.

2.5 Results

PassNet was tested on different scenarios, depending on the similarity of conditions to the match used for training. PassNet shows significantly better performance compared to the random prediction method as well as other baselines, as evidenced by the accuracy, F1 score, and precision metrics.

	1H of AC Chievo vs. Juventus FC				
	PassNet	ResBi	Baselines		
	YI = .003	YI = .0064	Random	Most	Least
<i>Acc</i>	70.00	63.37	49.99	65.08	34.92
F1	63.73	58.17	39.59	0.0	51.77
<i>Prec</i>	55.15	48.38	32.82	0.0	34.92
<i>Rec</i>	75.48	72.95	49.89	0.0	100
<i>PrecNo</i>	83.60	80.05	67.13	65.08	0.0
<i>RecNo</i>	67.06	58.23	50.04	100	0.0



(a): Real(red) and predicted(blue) labels.

In summary the results show that PassNet makes a significant contribution to pass detection and has good classification results, regardless of the video conditions for test and training sets. It is also in good agreement with sets of passes annotated by human operators of a leading data collection company, as mentioned by the authors of the paper.

2.6 Conclusion and Future improvements

Ultimately PassNet is a good model that shows promise for generalization in other football actions and even sports. Although it is still very much reliable on manual annotation by humans, it exceeds expectations considering the vastly different video conditions on the training set and inherent challenges from live-streamed data.

There are however areas for improvement:

- ▶ **Fixed camera views** to reduce noise from broadcast videos, enhancing the consistency and reliability of detections.
- ▶ **Custom YOLOv3 Model:** Training a custom YOLOv3 model for players and balls can improve accuracy and adaptability to sport-specific dynamics.
- ▶ **Player Identity Recognition:** Integrating capabilities for player identity recognition, such as jersey number detection, would enrich the analysis by associating actions with specific players.
- ▶ **Extension to Other Events:** Expanding the model's scope to annotate other critical events beyond passes could provide deeper insights into overall gameplay dynamics.
- ▶ **Accuracy Assessment:** Developing mechanisms to differentiate between accurate and inaccurate passes.

In conclusion, while PassNet demonstrates strong classification capabilities for passes, further advancements are necessary to generalize the model and enhance its capabilities and applications in sports analytics.

3. A Novel Framework for Fine Grained Action Recognition in Soccer

A challenging aspect of video-based action recognition lies in accurately identifying fine-grained actions, particularly in sports such as football where dynamic camera views and multiple players are involved. The complexity increases when the rapid movements, varying angles, and occlusions which are common in live sports broadcasts are taken into account.

To address these challenges, the authors propose a framework for event-based action recognition. This framework processes a video input and outputs the same video with individual actions tagged and highlighted within bounding boxes.

3.1 Framework overview

The authors are particularly interested in the recognition of different soccer actions like passing, shooting, heading etc. . The framework consists of primarily two modules, Event Detector and Action classifier, specifically:

- ▶ **Event Detector Module:** The Event Detector Module is responsible for identifying and localizing player actions using ball and player tracking and then segments the video into manageable clips centered around the detected actions ensuring that each clip contains the complete context of the action.
- ▶ **Action Classifier (GAWAC):** The Action Classifier, named Gaussian Weighted Action Classifier (GAWAC), is designed to provide precise classification of football actions. It employs a 3D Convolutional Neural Network (3D CNN) with Gaussian Weighted filters to enhance the accuracy of action recognition, emphasizing on the most relevant features while diminishing the influence of less important ones.
- ▶ **Output:** The final output of the framework is a set of tagged video clips with clearly bounded action boxes, The combination of tagged actions and bounded boxes transforms the raw video footage into an enriched format that is highly informative and easy to analyze.

3.2 Soccer-8k Dataset

The authors created the publicly available Soccer-8k dataset, consisting of 7942 labeled action clips. They used full match videos of LaLiga and Champions league games (Full HD definition) available on YouTube. These videos were processed by the Event Detector Module, which identified and extracted one-second video clips focusing on events where

the ball is in play. This allowed them to pinpoint the moments they wanted to focus on, such as when the ball is released or gathered in open play.

However, there are some challenges when creating a reliable Dataset:

- ▶ The ball often got hidden behind players
- ▶ Fast movement can make relevant objects(ball, players) blurry, making them hard to track
- ▶ Changes in camera angles and zoom levels complicate the tracking and posed a serious challenge.

Despite these challenges, the team was able to built a reliable Event Detector Module containing only the player actions with the ball.

After disregarding some frames (e.g. replays and close ups) they focused on identifying the Pivot Frames. A frame is tagged as a Pivot frame if the ball is in contact with a player in that frame but not in the preceding or succeeding frames. These are classified into:

- ▶ Type 1: Action occurs in succeeding frames (receiving the ball)
- ▶ Type 2: Action occurs in preceding frames (releasing the ball)

The surrounding frames are then extracted while the span is usually set to 24 frames.

After generating the clips, a team of football experts labels them into 8 different categories, while controversial clips are discarded(<50% agreement)

The dataset comprises 7942 labeled action clips across six classes:

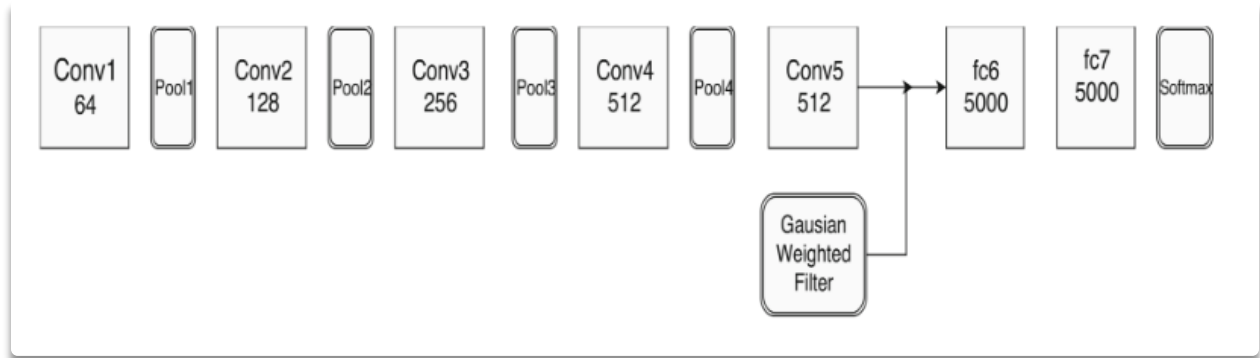
- ▶ **Short pass (2622 samples):** A low pass intended for a teammate's control.
- ▶ **Long pass (670 samples):** A long-distance pass.
- ▶ **Header (87 samples):** Using the head to pass, shoot, or clear the ball.
- ▶ **Trapping (1946 samples):** Controlling and intercepting the ball.
- ▶ **Dribbling (1991 samples)**
- ▶ **Turning (626 samples):** Changing direction with the ball.

After the dataset was finished the team run some experiments with 70-30 split for training and testing . They used data augmentation methods so all the classes are equally distributed(e.g. Header class, only had 87 samples).

3.3 GAWAC Architecture

GAWAC (Gaussian Weighted Action-Centered) is a Convolutional Neural Network (CNN) specifically designed for action recognition in videos. It uses a 3d kernel to capture spatial and temporal dependencies in video frames and also 5 Convolutional Layers, 4 Max-Pooling layers, 2 fully connected layers and 1 SoftMax Output Layer, specifically:

- **Convolutional Layers** have a kernel size of $3 \times 3 \times 3$ and $2^{5+n}, n = \text{The order of each Layer}(1 - 4)$ filters (512 also for layer 5)
- **Max-Pooling Layers** have a pooling size of $1 \times 2 \times 2$ (reduces spatial dimensions) For the first one, and $2 \times 2 \times 2$ for the other 3 (reduces both spatial and temporal dimensions)
- After the last Convolutional Layer, a **GWF (Gaussian distributed Weight Filter)** is applied to emphasize information at the center, filtering out irrelevant data.
- Finally, a **SoftMax Output Layer** is applied to produce a probability distribution across all eight possible action classes.



3.4 Results

The team tested the performance of three models on the Soccer-8k dataset, using confusion matrices and F1 scores. The Gaussian filter with a variance of 0.5 achieves the highest F1 score as it does not capture unnecessary information. Although the models were expected to perform equally across all the different classes, the limited number of original samples in the dataset may have resulted in inadequate training.

Model	F1 Score
C3D	46.02
Inception + LSTM	41.88
GAWAC with $\sigma^2 = 1.0$	55.72
GAWAC with $\sigma^2 = 0.5$	62.75
GAWAC with $\sigma^2 = 0.2$	52.8

The GAWAC model captures high F1 Scores, due to the localization with the gaussian filter

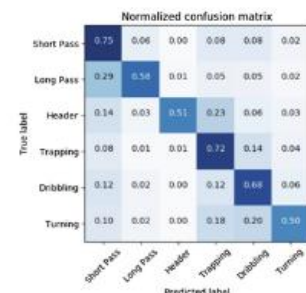


Fig. 8. Confusion matrix of GAWAC trained on Soccer-8k, $\sigma^2 = 0.5$

3.5 Data Augmentation

Models require a large amount of data to generalize well on unseen samples. Data Augmentation helps by increasing the variability of examples in datasets. We can use standard procedures like rotating, cropping, zooming etc., but due to the popularity of Deep Neural Networks and Convolutional Neural Networks there has been a growing interest to achieve better results by modifying the networks architecture and tailoring specific methods to particular datasets and models. For example, in the 8k Dataset the uneven class balance can result in a model that is bad in generalizing on classes with small sample size. Although the authors of this paper did not specify the data augmentation methods they used, we can refer to a method that can effectively handle temporal dependencies between frames:

- **Variational Autoencoders (VAEs):** VAEs can be trained on existing video clips from the Soccer-8k dataset to generate new, synthetic clips similar to the original data. 3D Convolutional Layers are needed to capture both spatial and temporal features. The VAE is very good in image denoising and in this specific case it will be a useful “ally” against motion blur and temporal noise contained in the sample images. The VAE is composed of two components, an Encoder, and a Decoder. The Encoder compresses the noisy input video sequence into a lower-dimensional latent space. It outputs the mean and standard deviation of the latent distribution, while the Decoder reconstructs the input information from this latent representation. This latent space is a lower-dimensional manifold of the input data. Then, synthetic data are generated, interpolating the values of the latent space, and decoding them. In practice, for this example we need high-quality clean training data which can be captured from an independent camera without the live stream format. We can then introduce synthetic noise (e.g. temporal noise) to the video sequences and train the VAE on these pairs(encoder->noisy sequences, decoder->clean sequences), after training the VAE on our dataset we can use the decoder to generate new synthetic data samples. For the synthetic noise if we want to replicate the effects of motion blur, we can use a Gaussian blur filter and for the temporal noise we can add random noise (e.g. Gaussian noise). So VAEs are a powerful tool that can both denoise our video frames and generate synthetic data that resemble real actions, especially for the classes with less samples.

3.6 Conclusion and Future improvements

The result of this paper was a very useful dataset containing ~8000 samples and with minor adjustments, the framework can be generalized to other multi-player sports. In consideration to the first paper this has much better-quality training data (more fps, Full HD), however improvements are crucial to extend it and make it a very useful video analysis tool. Areas of improvement, are:

- ▶ Classification of actions into subtypes (even finer action recognition), enhancing overall accuracy and specificity.
- ▶ Better techniques and experimentation with data augmentation, so that we can have evenly distributed data across all categories, diversifying the dataset and improving model generalization.
- ▶ The Addition of data denoising to mitigate artifacts such as motion blur and temporal noise. This enhancement will ensure cleaner and more consistent input data.
- ▶ Address the limitations associated with using broadcast footage by exploring options for live recording of matches. This approach can provide more controlled and high-quality data, reducing variability and enhancing the reliability of training datasets.

4 A Paradox

While the goal of automation is to reduce human intervention and improve efficiency, achieving this goal relies on the foundational step of creating high-quality training data, but also on updating and expanding datasets as we encounter new challenges or requirements.

Automation aims to replace manual tasks, but first it depends on manual efforts to create the data that makes automation possible, as we have seen on the papers mentioned above.

5 Closing statement

The ongoing research and advancements in video analysis represent a continuous effort to propose Cost-Effective solutions that will be used by small and big clubs alike. By automating tasks, we mitigate human error, save time, and optimize resource, a crucial step towards enhancing operational efficiency across sports. In the broader context, the field often remains underestimated. While sports are synonymous with entertainment and

commercialization, the solutions offered by video analysis are pivotal for injury prevention, performance evaluation, and tactical analysis. This research not only serves sports but also contributes valuable insights that generalize to other fields within computer vision.

The papers we have explored represent just a fraction of video analysis for sports and the importance of this research is self-evident.

The future of video analysis in sports is incredibly promising. Now IoT devices and wearable technology is also in the picture providing deeper insights into player health metrics and increasing the data available. With each breakthrough, we move closer to a more efficient sporting environment, where teams of all sizes can harness the power of data-driven insights to elevate their performance.

Whatever the future may hold, we are excited.

References

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